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OVGrasp: Target-oriented open-vocabulary robotic grasping in clutter☆

Xiaomei Zhang, Hanyue Ling, Xiao Huang, Qiwen Jin, Jiwei Hu *

School of Information Engineering, Wuhan University of Technology, Wuhan 430070, China

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ABSTRACT

Robotic recognizing and grasping of novel-category objects in cluttered environments presents a significant challenge due to occlusions and complex object arrangements. In addition, the ability to integrate language instructions is crucial for obtaining target object. In this work, we propose OVGrasp, an open-vocabulary grasping framework that seamlessly integrates vision and language to enhance robotic manipulation capabilities. Our approach leverages a unified integration of pretrained vision-language and grasping models, incorporates cross-modality alignment modules to enhance visual-linguistic perception, and uses a multi-scale voxel based point cloud representation for precise grasp-pose estimation in cluttered environments. By jointly modeling vision, language, and action, OVGrasp eliminates the reliance on predefined object labels and handcrafted rules, enabling more adaptable and efficient grasping. Extensive experiments in both simulation and real-world settings demonstrate that our method can achieve better task success rate by less times of motion, outperforming state-of-the-art methods under open-vocabulary language instructions in cluttered scenarios.

1. Introduction

Target-oriented grasping in clutter is an essential task for robotic manipulation, and has been explored for decades [1,2]. Common practices to specify the target object are to provide a target object image or collect a set of grasp demonstrations [3]. But providing such information is often infeasible for users, especially in open-world applications, such as open-vocabulary objects recognition and grasping.

An common idea is to replace visual command with natural language to assign the target object. Recent advancements in the field of visual-linguistic robotic grasping have facilitated human-guided interaction with robots through language. Large language models (LLMs) and vision-language models (VLMs) provide particularly promising tools for achieving more intelligent and adaptable grasping systems [4,5] by encapsulating extensive prior knowledge from broad data [6,7]. However, these works learn raw image-based policy with lots of simulated demonstration data. The cluttered scene may further increase the data requirement and sim2real difficulty.

The increasing demand for intelligent robots in diverse real-world applications, robots must adapt to recognize and grasp objects of different categories in complex environments [8]. Existing methods are limited to recognizing predefined (seen) categories [9], hindering their ability to handle novel-category (unseen) objects in real-world scenarios where listing all possible categories is impractical. This exposes limitations in current robotic perception and grasping methods. Hence, it is highly necessary to introduce an algorithm for detecting and grasping novel category objects in the field of robotics.

Robotic recognizing and grasping of novel-category objects in cluttered environments presents a significant challenge due to occlusions and complex object arrangements. VLG [2] primarily addresses the limitations caused by the separation of visual grounding and grasp planning in traditional methods. By jointly modeling vision, language and action, it reduces reliance on handcrafted rules. However, VLG suffers from poor performance in grasping small objects, as its single-scale feature extraction struggles to capture fine details and is easily affected by background noise in cluttered scenes, leading to insufficient grasp localization accuracy. OVGNet [10] proposes a unified visual-linguistic framework to handle novel-category objects. By aligning cross-modal features and incorporating positional information into language templates, it enhances contextual association in target recognition. However, in extremely densely stacked scenes, it suffers from insufficient grasping accuracy for small objects and fails to fully exploit multi-scale spatial information, limiting its ability to accurately estimate grasp poses and adapt to objects with complex geometries.

In this work, we propose OVGrasp, an open-vocabulary grasping framework that seamlessly integrates vision and language to enhance robotic manipulation capabilities. As shown in Fig. 1, our approach features a unified vision-language model and grasping model for grasping both base and novel objects; it incorporates cross-modality attention and alignment modules to enhance visual-linguistic perception for

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* Corresponding author.
E-mail address: hujiwei@whut.edu.cn (J. Hu).

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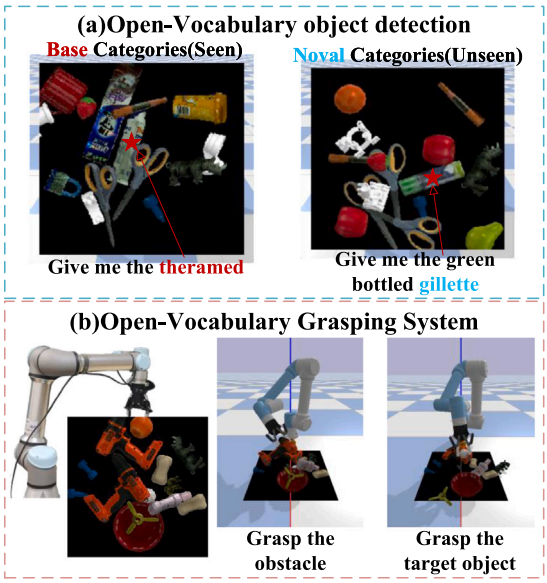

Fig. 1. The diagram of open-vocabulary grasping. Objects and fonts in red and blue respectively indicate the base and novel categories. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

novel objects within the visual-linguistic module. For precise grasp pose estimation in cluttered environments, it employs a multi-scale voxel-based point cloud fusion strategy to fully exploit spatial information across scales. By jointly modeling vision, language and action, OVGrasp eliminates the reliance on predefined object labels and handcrafted rules, enabling more adaptable and efficient grasping. Extensive experiments in both simulation and real-world settings demonstrate that OVGrasp achieves a high success rate, outperforming state-of-the-art methods in cluttered scenarios and under diverse language instructions. Our framework represents a step forward in enabling robots to generalize effectively to novel objects and commands, enhancing their real-world applicability.

Specifically, the primary contributions of our proposed network are as follows:

- We propose a novel task-oriented Open-Vocabulary robotic grasping framework which leverages both pre-trained visual-language model and grasping model, significantly enhancing the robot's ability to grasp novel category objects.
- We utilize the fine-tuning techniques Clip-Adapter to improve novel object perception and alignment module LGFA to enhance image-text feature consistency. We introduce a multi-scale voxel point cloud fusion strategy to enable accurate grasping capability in clutter.
- Our system is evaluated on a series of scenarios with seen and unseen objects and language instructions in simulation and real-world settings, of which the results validate the effectiveness and generalization.

The remaining sections of this paper are organized as follows. Section 2 reviews related work on open-vocabulary visual grounding and vision-language robotic grasping. Section 3 presents the proposed OVGrasp network architecture. Section 4 reports detailed experimental results. Section 5 provides a discussion of the strengths and limitations of our approach. Finally, Section 6 concludes the paper.

2. Related work

2.1. Target-oriented grasping in clutter

Robotic grasping in cluttered environments has been an active research area for decades. Due to the complexity of occlusions and the diversity of objects, it is very difficult to capture in a cluttered environment. Traditional methods rely on generating 6-DoF grasp proposals of a clutter of every objects, and then using mature robot path planning and control methods to execute the grasping [11]. GraspNet [12] has constructed a dataset containing one billion grasp poses and provides a baseline for detecting grasp poses. Furthermore, Lu et al. [13] propose the FGC-Grasp which integrates force-closure with measurements of flatness, gravity, and collision, to improve grasp stability. However, these models struggle with generalization and robustness in diverse, unstructured environments. Another line of works studies target-oriented grasping by assigning an image of the target object [14], or providing a set of grasping demonstrations. Gergely et al. [15] utilization of learned implicit representations of visual and geometric features derived from a pre-trained NeRF train a grasp evaluation model from a small number of demonstrations to address the problem of robotic grasping of known and unknown objects. However, they often need a lot of data to be collected and labeled, which makes them less useful in a variety of situations.

Recently, deep learning methods based on Convolutional Neural Networks (CNNs) and Reinforcement Learning (RL) have made significant progress in exploring target-oriented grasping in cluttered environments. [16–19] enable systems to identify and estimate the locations of obstacles that may hinder the grasping of the target object. By employing a reinforcement learning mechanism, the system will first grasp and manipulate other surrounding obstacles, effectively clearing a path to ensure the target object can be successfully reached and grasped. [4,20,21] conduct some pre-grasping actions such as pushing where the target object may be partially or fully occluded by other objects. By pushing the nearby obstacles, the system can create a clear path to the target object, reducing interference and minimizing the chances of accidental collisions. However, they often need an image of the target object or assume that a target object mask in the scene is given, which is often infeasible, especially in open-vocabulary applications, such as tabletop sorting and household assistance.

2.2. Open-vocabulary vision grounding

Visual grounding grounds semantic concepts to image regions [22]. Specifically, the task can be divided into: (1) phrase localization [23] that grounds all nouns in the sentence; (2) referring expression comprehension (REC) [24,25], that only grounds the referent in the sentence, typically providing a target bounding box as the result. In the latter, the referent is labeled not with a class name but with freeform natural language describing instance attributes, positions, and relationships with other objects or backgrounds. While this task significantly expands the range of concepts that models can ground, the vocabulary remains closed-set.

Open-Vocabulary learning (OVL) [26] is proposed to expand the range of vocabulary and comprehension. It has been successfully applied to multiple tasks, e.g., segmentation, detection, video understanding, scene understanding, etc. Previous works mainly involved two aspects: Open-Vocabulary Detection (OV-D) [27,28] and Open-Vocabulary Segmentation (OV-S) [29,30]. Similarly, the model trains on base classes and inferences on both base and novel classes like zero-shot learning. An ideal way to achieve OV-D and OV-S is to scale the small-scale grounding datasets to web-scale datasets. However, the laborious labeling cost is non-negligible.

Unlike the OV-D and OV-S tasks, Open-Vocabulary Visual Grounding (OV-VG) [31] aims to detect and localize novel-category objects referenced by natural language, thus establishing a more comprehensive

correspondence between language and vision. During the pre-training phase, the model may be exposed to a broader vocabulary, which may include both base and novel classes. To the best of our knowledge, current visual grounding methods, such as VLTVG [32], face significant challenges when addressing the open-vocabulary problem. Existing models built upon the OV-D framework primarily focus on object detection, which often leads to attempts at predicting all objects in an image [33]. However, since language descriptions do not always align precisely with specific image regions, these models often struggle with accurately distinguishing the intended target from other detected objects.

2.3. Vision-language robotic grasping

In recent years, the convenience and appeal of natural language has propelled research into language-conditioned robotic manipulation. By integrating visual and language information [34], multimodal learning has enabled robots to understand human language instructions and generate accurate grasp pose [35–38]. Language-guided grasping typically involves two stages: first, obtaining a target object's bounding box or segmentation mask, then feeding it into a grasping pipeline [39]. Some researchers have explored end-to-end approaches for generating grasp poses in Referring Grasp Synthesis (RGS) [35,36], while others have focused on predicting grasp poses via affordance maps in Referring Grasp Affordance (RGA) [9], bypassing the need for object bounding boxes or segmentation masks.

With the development of large language models and multi-modal models [40], vision-language guided robotic grasping has become a cutting-edge research topic. Vision-Language Models (VLMs), trained on vast image-text datasets, excel at visual reasoning tasks and have been popularly applied to diverse robotics problems. Models such as CLIP [6] and CLIPort [41] have improved task performance. Xu et al. [2] access the probability of each grasping pose by jointly modeling action, language, and vision. Tang et al. [36] design a task-oriented grasp method to address the challenge of task grounding in addition to object grounding. Lu et al. [5] proposed a Vision Grounding dataset called RoboRefIt to train visual-linguistic model and developed interactive 6-Dof grasp policies for cluttered scenes.

Additionally, Large Language Models (LLMs) can generate high-level robotic execution plans based on task inputs and environmental context [42]. Voxposer [43] and GraspGPT [4] have demonstrated how LLMs can generate task-relevant actions and grasping strategies. However, a recurring challenge with LLMs is their tendency to hallucinate, generating plans that are not physically feasible [44]. To enhance robustness, LLMs require real world grounding, which can be achieved through feedback from the environment [42,45], integration with visual perception systems [46,47], or human-in-the-loop interventions like question-answering [48]. Models like ManipVQA [38], RoboScript [49], CoPa [50] use vision language models and contextual information to improve the performance of grasping tasks. Despite the successful integration of visual-linguistic models into the robotic grasping task, allowing for effective manipulation of target objects within the base category, the challenge of grasping novel objects persists, they also have to address the challenges of sim2real.

3. Proposed method

In this section, we present the detailed framework of our proposed method, as shown in Fig. 2. Given a set of object-centric bounding boxes and a text instruction, we extract visual and language features using pre-trained CLIP encoders, fine-tuning the image encoder with CLIP-Adapter for enhancing perception of novel categories and small objects. We then obtain positional features by embedding the bounding box centers. To align features across visual, textual, and positional modalities, we introduce cross-modal attention mechanisms, where image-to-text and text-to-image attention generate visual-language features, image-to-position and position-to-image attention obtain visual-positional features. Then we encode grasp poses into spatial features using the pre-trained GraspNet [12], ensuring that grasp pose representations interact effectively with both visual and language features. Additionally, we propose a cross k-NN (k-Nearest Neighbor) attention transformer, which selects the top relevant vision, language, and positional features to guide grasp prediction. To refine target object reference, we incorporate Language-Guided Feature Attention (LGFA), ensuring that visual features are focused on the specific regions described in the instruction. Our framework also leverages multi-scale point cloud reconstruction and fusion to enhance spatial awareness and grasp pose estimation. The policy and critic networks, trained via reinforcement learning, jointly optimize grasp selection, leading to more efficient and adaptive grasping strategies. During inference, our method effectively generates a sequence of grasp poses to remove obstacles and ultimately grasp the target object.

3.1. Visual-linguistic perception module

Rather than directly applying CLIP to the raw image of a cluttered scene, we propose processing object crops to obtain a set of object bounding boxes as inputs, denoted as $v \times N$. Due to object occlusion in clutter and the detection uncertainty, the total number of objects is not equal to N in most cases. Given a language expression l, we encode the cropped images and text into the CLIP image backbone $f_v(\cdot)$ and text backbone $f_l(\cdot)$, obtaining the image feature v_v and text feature v_l, respectively. Meanwhile, we incorporate positional information by encoding the center coordinates of the bounding boxes as positional embeddings v_p.

To better adapt CLIP to the open-vocabulary robotic grasping task, we employ a variant named CLIP-Adapter, as shown in Fig. 3, which fine-tunes only the visual adapter of CLIP to improve performance. Rather than modifying the entire backbone, we introduce a small number of additional learnable bottleneck linear layers to the CLIP image branch while keeping the original backbone frozen during fine-tuning. Specifically, the visual adapter A_v consists of two linear transformation layers, where the first layer projects the original image feature f into a lower-dimensional space via $\mathrm{ReLU}(f^T W_1^v)$, and the second layer maps it back to the original dimensionality using W_2^v. This lightweight adjustment mitigates the risk of overfitting, especially in few-shot scenarios. Additionally, to effectively balance the trade-off between generalization and task-specific adaptation, we leverage CLIP-Adapter's residual connections to dynamically integrate fine-tuned features with the original representations from CLIP's backbone. The new image feature f^* is computed as:

$$f^* = \alpha \cdot A_v(f)^T + (1 - \alpha) \cdot f \tag{1}$$

where α acts as a residual ratio to adjust the contribution of fine-tuned knowledge and pre-trained features. This allows us to retain the strong generalization ability of CLIP while improving its grasping-specific recognition. Furthermore, as the optimal value of α varies across datasets. For example, fine grained tasks often require larger α to prioritize new knowledge, while generic tasks perform better with smaller α manually tuning it can be both time-consuming and labor-intensive. Thus, we refine our approach by learning α as a trainable parameter via a hypernetwork, enabling adaptive weighting of fine-tuned and pre-trained features for improved open-vocabulary detection.

During training, the original CLIP backbone remains frozen, with only the visual adapter A_v and learnable α optimized via cross-entropy loss, computed from predicted target category probabilities. We use a batch size of 32, initial learning rate of 1×10^{-5}, and set the bottleneck layer's hidden dimension to $D/4$ for a balance of semantic preservation and efficiency. This setup leverages residual connections to retain pre-trained knowledge, mitigating overfitting in few-shot scenarios.

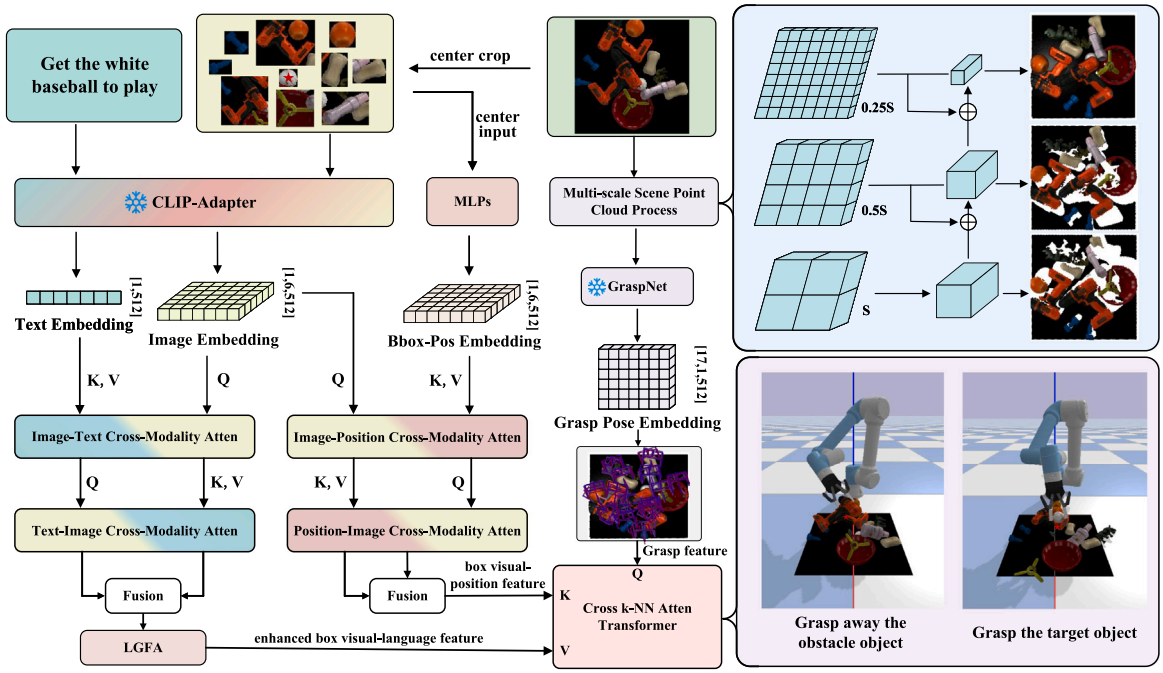

Fig. 2. The overview of OVGrasp. The visual-linguistic locate perception system locates the target object referred by natural language. The grasping system generates grasping pose for the target object. LGFA represents the language guided feature attention module. Enforcing the image features to focus on referring expression regions.

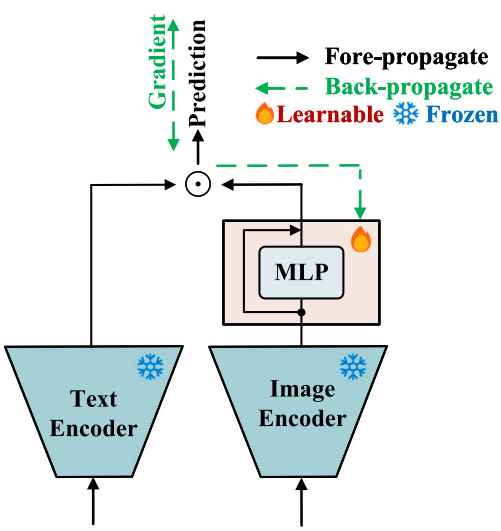

Fig. 3. CLIP-Adapter. Based on CLIP, fine-tuning is performed using feature adapters on the visual branch.

Next, we introduce a cross-modality attention mechanism to deeply integrate these multi-modal features. Specifically, we iteratively traverse and fuse the outputs of each layer from the image and text encoders, as well as the image and position encoders. After fusing the outputs of the current layer, the results are used to update the inputs of the next layer, enabling further integration of visual, textual and positional modalities. For example, we define $n = \min(N_v, N_l)$, where N_v and N_l represent the number of layers in the image and text encoders, respectively. The updated visual features v'_v can be expressed as:

$$v'_v = \text{concat}\left(\mathcal{P}_{v\to l}\left(\mathcal{P}_{l\to v}\left(v_v(i), v_l(i)\right)\right), 0 \le i \le n\right) \tag{2}$$

$\mathcal{P}_{v\to l}(\cdot)$ and $\mathcal{P}_{l\to v}(\cdot)$ mean the image-to-text and text-to-image fusion. $v_v(i)$ and $v_l(i)$ are ith layer of image and text encoder. Through this process, we get the enhanced box visual feature v'_v and enhanced language feature v'_l. Simultaneously, the updated positional features v'_p are formulated as:

$$v'_p = \text{concat}\left(\mathcal{P}_{v\to p}\left(\mathcal{P}_{p\to v}\left(v_v(i), v_p(i)\right)\right), 0 \le i \le n\right) \tag{3}$$

$\mathcal{P}_{v\to p}(\cdot)$ and $\mathcal{P}_{p\to v}(\cdot)$ represent the image-to-position and position-to-image fusion. $v_v(i)$ and $v_p(i)$ are ith layer of image and position encoder. So we obtain the enhanced box visual feature v''_v and enhanced box position feature v'_p. Subsequently, each visual feature v'_v is fused with language feature v'_l to generate box visual-language feature, while each visual feature v''_v is fused with position feature v'_p to generate box visual-position feature.

Finally, to align the consistency between vision-language features, we introduce the Language-Guided Feature Attention (LGFA) mechanism to further refer to the target object. The LGFA model is based on multi-head attention, where the query is enhanced image feature v'_v, while the key and value correspond to the enhanced text embedding v'_l. Through multi-head attention, the enhanced image features are aligned with the enhanced text embeddings to generate a semantic map v'_s. Subsequently, we use linear projection and L2 normalization for mapping v'_v and v'_s to the same space, expressed as $\hat{v}'_v$ and $\hat{v}'_s$, then we calculate the attention score for each point x, where α and σ are learnable parameters:

$$S_x = \alpha \cdot \exp\left(-\frac{\left(1 - \hat{v}'_v(x)^{\mathrm{T}}\hat{v}'_s(x)\right)^2}{2\sigma^2}\right) \tag{4}$$

After obtaining the language-guided image feature attention scores about semantic relevance, we can mathematically get the most relevant area in the image according to text information. We take the dot product result of the above-mentioned score and v'_v as a new enhanced box visual-language feature v'''_v to fusion model, shown as

$$v'''_v = \beta \cdot v'_v \cdot S_x + (1 - \beta) \cdot v'_v \tag{5}$$

where β denotes the balance parameter, which is empirically set to 0.6.

3.2. Grasping pose detection module

This module is designed to guide robots in generating grasp poses for target objects within cluttered environments. During the experimental process, we observed that generating an accurate grasping pose for smaller objects is challenging. Therefore, we utilize a multi-scale voxel scene point cloud to extract the target point cloud. Given multiple initial point clouds $\mathcal{P} = \{P_1, P_2, \ldots, P_N\}$, we first define a set of multi-scale voxel sizes $V = \{v_0, v_1, v_2\}, \quad v_i = \alpha^i v_0$, where v_0 is initial voxel size, α controls the ratio of voxel refinement. In our network, we employ a three-layer scale fusion strategy with the values $\{v_0, v_0/2, v_0/4\}$. Next, we apply statistical filtering to the input point cloud P_i to remove outliers and perform voxel downsampling at multiple scales, progressively refining the point cloud structure $P_i^{(v_i)}$ from a global to a local level. At each scale, point cloud registration is conducted using a nearest-neighbor search and a Point-to-Plane matching strategy for precise alignment. A dynamic convergence threshold is set to ensure registration stability. For low-density regions, larger voxels are used for coarse fusion, while high-density regions are processed with smaller voxels for fine-grained optimization, balancing computational efficiency with detail preservation. Finally, we integrate multi-scale point clouds:

$$P_{\text{fused}} = \bigcup_{i=0}^{2} \text{Merge}\left(P_{\text{fused}}, P_i^{(v_i)}\right) \tag{6}$$

to leverage complementary information across resolutions and enhance the completeness and accuracy of the point cloud representation on small objects.

Subsequently, we feed the processed point cloud into pre-trained GraspNet to generate a set of 6-DOF grasping poses for the entire scene. To enhance the success rate of grasping and filter out unsuitable postures, we apply the grasp score obtained from GraspNet and grasp angle as thresholds to filter grasp poses. Following this, in the process of selecting the target grasp pose, we compute the Euclidean distance to compare the distances between the center of each bounding box and the center of each grasp pose. The grasp pose with the shortest distance to the bounding box center is selected as the final target grasp pose, ensuring precise and stable object acquisition.

3.3. Cross-modal joint modeling

In order to jointly capture features to obtain target object grasping poses marked by language instructions, we fuse multi-modal features through a cross-attention transformer. However, the standard dense self-attention mechanism, which computes attention across all image patches, can be inefficient. This approach often neglects the locality of image patches and may include irrelevant information, such as cluttered backgrounds or occlusions, resulting in slower training and potentially degraded performance. To address these issues, we introduce cross k-NN attention transformer, which refines the traditional attention mechanism by focusing only on the top-k most relevant tokens. Instead of involving all tokens in the attention matrix computation, we select the top-k most similar tokens from the key set for each query. This reduces noise, improves computational efficiency, and enhances the relevance of the attended features. Specifically, we calculate the dot product between queries and keys, then select the top k elements row-wise for softmax computation. The procedure can be formulated as:

$$\hat{V}_{\text{knn}} = \text{softmax}\left(\tau_k\left(\frac{QK^T}{\sqrt{d}}\right)\right) \cdot V \tag{7}$$

where τ_k denotes the row-wise top-k selection operator:

$$\left[\tau_k(A)\right]_{ij} = \begin{cases} A_{ij} & A_{ij} \in \text{top-k(row } j) \\ -\infty & \text{otherwise} \end{cases} \tag{8}$$

the only parameter for k-NN attention is k. Generally speaking, the value of k approximates half of the total number of tokens n. In our network, we set $k = 100$.

We first fuse the N enhanced box visual features v_v' with the enhanced language feature v_l' using element-wise multiplication, obtaining N corresponding visual-language features accordingly. Additionally, refine these features through Language-Guided Feature Attention (LGFA) to derive language guided box visual-language features v_v''. Then, we generate K grasp pose features by MLP. Given that our objective is to achieve joint modeling of vision, language, and grasp features, we set the K grasp pose features as queries and the N enhanced box visual-language features as values. To incorporate spatial features (i.e. grasp features) into the visual-language representation, we further utilize the box visual-position features as keys. To be specific, 3D positions of the centers of bounding boxes are projected into a nonlinear space, followed by a MLP to encode the box position embeddings. These embeddings v_p' are then fused with the enhanced box visual features v_v'' through element-wise addition. Finally, the cross k-NN attention transformer generates K vision-language-grasp features, which are then fed into the decision network for grasp execution.

3.4. Implementation details

By jointly modeling vision, language, and grasp poses while integrating cross-attention with grasp queries, our system enables language-driven grasping without requiring object labels, visual grounding, or handcrafted planning rules. This approach enhances adaptability to diverse language instructions, particularly for open-vocabulary scenarios where novel or previously unseen commands may arise. Our system is trained with five language templates: "Give me the keyword", "I need a keyword", "Grasp a keyword object", "I want a keyword", "Get a keyword to ...", where keyword can be a concrete label (e.g. banana, apple), or a general label (e.g. fruit), or the attribute of shape (e.g. round), or object function (e.g. eat, play). The network parameters of CLIP and GraspNet will be fixed during training.

4. Experiments and results

In this section, we assess the effectiveness and rationality of our proposed framework for the open-vocabulary grasping task in cluttered scenes. This evaluation includes the success rates for grasping in both base and novel scenes, as well as the number of successful motions.

4.1. Dataset

To demonstrate that our method is effective for the task of language-conditioned grasping in clutter and evaluate the generalization performance of our network on unseen objects and language instructions, we used a benchmark dataset OVGrasping [10] specifically tailored for the open-vocabulary grasping (OVG) task. The OVGrasping dataset comprises 117 categories and 63,385 instances. Specifically, the dataset is divided into two categories: the base category consists of 68 classes with 51,857 instances, and the novel category comprises 49 classes totaling 11,528 instances. The base category is used for training our framework, whereas the novel category is only utilized to evaluate the performance of open-vocabulary. Additionally, to evaluate the performance within the base category, we randomly extract 10% from the base category as the test set with 4,830 instances.

4.2. Experiments setting

Training Policy: In this work, our system is trained using deep reinforcement learning with discrete Soft Actor-Critic (SAC) [51]. At step t, we define the cross features as the state s_t and the grasp pose selection as the action a_t. The policy is then defined as $\pi(a_t|s_t)$, and the critic outputs Q-function $Q(s_t, a_t)$. Both the policy and critic MLPs take K cross features as input and generate logits and corresponding Q values for K grasp poses. Notably, our policy and critic networks are designed to handle a variable number of actions by processing K cross features in parallel, each corresponding to a different grasp pose.

Training Schedule: Training a policy directly in a cluttered environment may cause the agent to struggle with balancing the removal of obstructing objects and focusing on grasping the target object. Since our ultimate objective is to grasp the specified target, we adopt a two-stage curriculum learning strategy. Stage slowromancapi@: Scattered Scene. In this stage, we train our policy in a scattered environment with 9 objects randomly dropped in the workspace. The goal is to guide the robot to focus on grasping the language-guided target object while penalizing attempts to grasp other objects or failed grasps. To increase the robustness of target discrimination, we introduce distractors in several configurations: visually similar objects with different colors, where the grasping conditions include explicit color instructions, and identical objects placed at different positions, where the grasping conditions contain spatial cues. The reward function encourages the robot to grasp the correct target within fewer attempts and penalizes grasping other objects, even if they are of the same category but not the intended target. The number of grasp attempts is limited to 8, and the episode terminates immediately once the target object is successfully grasped. The reward function is defined as:

$$R_g = \begin{cases} 1 & \text{success grasp} \\ -1 & \text{grasp others or failed grasp} \end{cases} \tag{9}$$

Stage slowromancapii@: Cluttered Scene. In this stage, our policy is trained in cluttered environments with 15 objects randomly dropped in the workspace, where the tight packing or occlusions degenerate the successful grasp of the target object. For each episode, the robot is guided to remove obstacles especially those located near the target object to facilitate its final grasp. The number of grasp attempts is limited to 8. To handle scenarios with multiple identical objects, the reward function applies a lower penalty for grasping obstacles close to the target, enabling the robot to use reward feedback to accurately distinguish and grasp the target object even in the presence of visually or categorically similar distractors. The reward function is as follows:

$$R_g = \begin{cases} 2 & \text{success grasp} \\ -\frac{\text{dist}}{\text{dist}_{\max}} & \text{success grasp other objects} \\ -1 & \text{failed grasp} \end{cases} \tag{10}$$

where $dist$ represents the 3D euclidean distance between the grasped object and the target object. $\text{dist}_{\max}$ denotes the diagonal length of the workspace. Dividing $dist$ with $\text{dist}_{\max}$ normalizes the value within $[-1, 0)$.

Evaluation Metrics: We evaluate the proposed methods through a series of test cases, each consisting of 15 runs, and measure performance with two key metrics: Task Success Rate and Motion Number. The Task Success Rate is the average percentage of successful task completions, where a task is considered successful if the robot picks up the target object within 8 action attempts in each run. The motion Number represents the average number of motions per task completion.

4.3. Simulation experiments

Our simulation environment is built in PyBullet [52], which involves a UR5 robotic arm, a ROBOTIQ-85 gripper for grasping objects, and an Intel RealSense L515 camera to capture images for the visual-language perception module and grasp pose detection module. We generate two types of grasping scenes to test evaluation indicators, of which one is used to test base categories on scattered scene and cluttered scene, and other one is used to evaluate novel categories or noval language template on cluttered scene. Our base and novel experimental scenarios are shown in Figs. 4 and 5. Our experiments encompass various tasks in clutter, such as grasping small round or irregular objects, retrieving items for eating or playing, and other specific requests, both including arrangements with seen and unseen objects. Arrangements with target objects involve 10 cases with adversarial clutter where the robot might need to grasp away other obstacles for target object grasping.

Table 1
Open-Vocabulary Simulation Results On Cluttered Scene with Ablation Studies. Annotation: bold black text indicates the best results.

Methods	Task success ↑		Motion Number ↓	
Arrangement	Seen	Unseen	Seen	Unseen
CLIP-Grounding	0.579	0.613	4.91	4.36
VLG [2]	0.693	0.727	4.22	3.73
OVGNet [10]	0.746	0.773	3.83	3.62
Ours	**0.879**	**0.893**	**3.48**	**2.62**
Ours w/LGFA	0.694	0.680	4.73	5.02
Ours w/k-NN	0.672	0.699	4.30	4.12
Ours w/multi-voxel	0.618	0.593	4.29	5.33
Ours w/CLIP-Adapter	0.685	0.660	4.86	5.12

To further verify the effectiveness of our framework, we compare our solution against state-of-the-art methods, Vision-Language Grasping (VLG) [2] and Open Vocabulary Grasp Network (OVGNet) [10]. Since the language provided by VLG can refer to multiple target objects, whereas in our setting, the language must refer to single target object. Therefore, we modify VLG's language templates by removing redundant objects or referencing additional descriptions, such as color and location, to ensure clear identification of the target object. At the same, in the original OVGNet setup, the scene terminates immediately after any object is grasped, with a maximum of 3 grasp attempts. To ensure a consistent evaluation, we increase the number of grasp attempts in OVGNet to 8, allowing the scene to continue until either a successful grasp is achieved or all attempts are exhausted. These modifications align both methods with our experimental setup, enabling a more rigorous and fair comparison.

Result: The results, summarized in Table 1, demonstrate that our system significantly outperforms the baselines in overall success rates and efficiency metrics. Specifically, our method achieved an average success rate of 0.879, with an average success step count of 3.48 in seen clutter case. These results indicate that our system not only excels in accomplishing grasp tasks but also operates with greater efficiency, requiring fewer steps for successful task completion. We also evaluated our system's performance in unseen clutter scenarios, where target objects are partially or completely occluded.

Among the 10 test examples, there are 5 groups of scene target objects that are not occluded, 4 groups of scene target objects that are partially occluded, and 1 group of scene target objects that are completely occluded. Each scene contains one or more unseen objects. All objects to be grasped are unseen. In scattered scenes, the average success rate of our method reached 0.926, and the average number of successful steps of grasping was 1.571. In cluttered scenes, our method achieved an average success rate of 0.896, and the average number of successful steps was 2.21. In particular, when the target object is occluded, our grasping success rate also reached 0.826, and the average number of successful steps was 3.027. We also found that in the experiments of scattered scenes, our framework tends to grasp a single scenes, with a minimum number of successful steps of 1.0, while in cluttered environments, our framework tends to grasp multiple scenes, with a minimum number of successful steps of 1.27. The above results effectively validate the grasping performance of our framework on new categories.

Fig. 4. Seen clutter cases in simulation. The target objects are labeled with stars. The blue text indicates the additional descriptions for same experimental setup. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

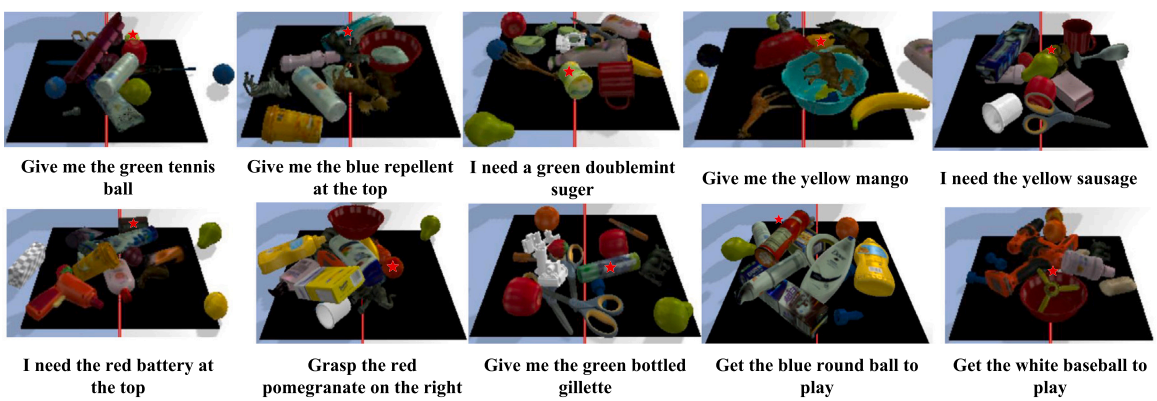

Fig. 5. Unseen clutter cases in simulation. The target objects are labeled with stars. Some severe obscured objects may not be visible.

Tables 2 and 3 present the detailed performance of our approach compared to the baseline methods on various tasks. Our method consistently achieves high success rates and lower average steps, demonstrating robustness and efficiency. Notably, in some partially occluded tasks such as "Give me the yellow mango" and "Get the blue round ball to play" our system outperforms other methods, indicating its ability to accurately identify and grasp target objects even in cluttered environments. However, in severely occluded environments, where the target object is not immediately identifiable, the robot must first grasp and move nearby obstacles to clear the scene. This increases the number of steps, as the robot must remove obstacles before successfully grasping the target object.

In tasks involving small objects, such as "Give me the green tennis ball", VLG achieves a success rate of only 0.5333, while OVGNet, despite reaching 0.8666, still requires a relatively large number of steps. In "Get the white baseball to play", where the target object is both small and occluded, VLG performs poorly with a success rate as low as 0.1333 (succeeding only twice in 15 attempts), primarily due to its reliance on single-scale feature extraction, which struggles to capture fine-grained details of small targets and is easily affected by background noise in dense scenes. OVGNet performs slightly better in this case, achieving a 0.2666 success rate, but still requires an average of 8.0 steps. The main reason is its lack of multi-scale fusion of 3D spatial information, which hinders accurate grasp pose estimation in stacked scenarios. In addition, OVGNet lacks fine-grained alignment between visual and language features, limiting its ability to distinguish semantically similar objects. For example, in the "Give me the yellow mango" task, it fails to differentiate between the target mango and visually similar yellow lemons (both yellow and oval-shaped, with overlapping visual features). In one experimental round, OVGNet repeatedly misidentified lemons as mangoes, resulting in a success rate of 0 for that round. These

Table 2
Detailed performance comparison of Average Success on the open-vocabulary grasping task, presenting our approach against baseline methods across various tasks.

Task	VLG	OVGNet	Ours
Give me the green tennis ball	0.5333	**0.8666**	**0.8666**
Give me the blue repellent at the top	0.7333	0.9333	**1.0000**
I need a green doublemint suger	**1.0000**	0.9333	0.9333
Give me the yellow mango	0.5333	0.6666	**0.8000**
I need the yellow sausage	**0.8666**	0.7333	0.6666
I need the red battery at the top	**1.0000**	**1.0000**	**1.0000**
Grasp the red pomegranate on the right	0.8000	0.8000	**0.9333**
Give me the green bottled gillette	0.9333	**1.0000**	**1.0000**
Get the blue round ball to play	0.7333	0.5333	**1.0000**
Get the white baseball to play	0.1333	0.2666	**0.7333**

Table 3
Detailed performance comparison of Average Success Step on the open-vocabulary grasping task, presenting our approach against baseline methods across various tasks.

Task	VLG	OVGNet	Ours
Give me the green tennis ball	3.33	2.20	**1.23**
Give me the blue repellent at the top	3.93	2.36	**1.87**
I need a green doublemint suger	**1.43**	3.58	2.36
Give me the yellow mango	4.66	3.40	**3.17**
I need the yellow sausage	**3.77**	4.64	4.90
I need the red battery at the top	2.57	2.35	**1.27**
Grasp the red pomegranate on the right	4.31	3.00	**2.14**
Give me the green bottled gillette	2.93	**1.53**	3.40
Get the blue round ball to play	4.71	5.11	**1.27**
Get the white baseball to play	5.63	8.00	**4.64**

Table 4
Testing performance on unseen language templates and the clutter scene is the same as before. According to the stacking degree around the target object, the test experiments are divided into unoccluded, partially occluded and severely occluded.

Occulsion	Target object	language template	Success rate	Motion number
No occulsion	green doublemint suger	Please pass me a ...	15/15	2.80
	green tennis ball	Can you pass me the ... to play	15/15	1.00
	red battery	I want to pick up the ... at the top	15/15	1.07
	red pomegranate	Can you please pick up the ... for me	13/15	2.31
	green bottled gillette	I want to use the ...	15/15	2.73
Partial Occulsion	blue repellent	Could you please bring me the ... at the top	15/15	1.66
	yellow mango	I want to eat the ...	12/15	3.58
	yellow sausage	I want to eat the ...	9/15	5.11
	blue round ball	Can you bring me the ... to play	15/15	1.33
Severe Occulsion	white baseball	Please pass me the ... to play	12/15	4.33

(a) (b)

Fig. 6. Settings of our real-world experiments. (a) Our experiment configuration. (b) The objects used in our experiments.

results are highly consistent with the inherent architectural limitations of the two baseline methods.

Furthermore, we also test our policy with novel language templates. As shown in Table 4, we evaluate our policy with 10 scenarios that include both seen and unseen objects, employing language templates that were not encountered during training. For example, "Please pass me the keyword", "Can you please pick up a keyword for me", "I want to eat the keyword". The results suggest that our policy is applicable for some unseen language templates in the open-vocabulary grasp task, which further confirms our claim on language generalization.

4.4. Real-world experiments

In this section, we evaluate our system in real-world settings. Our real-world platform involves a UR5 robot arm with a DH-ROBOTICS AG-95 gripper, and an Intel RealSense D435i capturing RGB-D images of resolution 1280 × 720. Object bounding boxes are generated by a pre-trained open-set detection model. As the grasping task is a downstream application of object detection, we employ YOLOv10, one of the current state-of-the-art detection models, to generate object bounding boxes, thereby ensuring high detection accuracy and robustness across diverse scenarios. Given that suboptimal detection results will inevitably propagate errors and degrade performance in subsequent grasping stages, our model integrates multi-modal feature fusion and multi-voxel point-cloud fusion to refine detection outputs and mitigate such errors as much as possible. To ensure consistency between simulation and real-world experiments, the AG-95 gripper is initialized in the closing direction, and its maximum stroke is set to 90% of its full range. This configuration provides a stable and repeatable starting state for all experiments. The test setting are shown in Fig. 6 and the test cases is shown in Fig. 7. We compare our methods with CLIP Grounding and VLG. Note that these methods are transferred from simulation to the real world without any retraining (see Table 5).

Fig. 7. Test cases in real world. Target objects are labeled with stars.

Table 5
Open-Vocabulary Real-world Results On Cluttered Scene. Annotation: bold black text indicates the best results.

Methods	Task success	Motion number
CLIP-Grounding	0.53	4.82
VLG [2]	0.69	5.48
Ours	**0.78**	**4.47**

Table 6
Ablations on varying the residual ratio α.

Ratio α	0.0	0.2	0.4	0.6	0.8	1.0
Average Success	0.660	0.780	0.873	0.893	0.833	0.807
Average Step	5.12	3.89	3.32	2.62	2.92	3.27

Result: Test results are reported in Fig. 7. Overall, our policy shows much better performance in task success rate, and is capable of generalizing to unseen target objects (i.e. strawberry, green repellent and white baseball). Although CLIP Grounding achieves a smaller motion number, it shows a task success rate low at 50~60%. For case 1 and 3, where CLIP grounds with high accuracy, CLIP Grounding can focus on the target grasping with less steps. However, when the target is occluded (case 4 and case 5), CLIP Grounding frequently fails to correctly identify the target. Meanwhile, VLG also exhibits certain limitations. Although VLG performs well in cases with clear object boundaries, its grasping performance degrades significantly for small objects or those with unclear edge details (case 2). When dealing with such objects, its reliance on region proposals leads to inaccurate localization, making it difficult to generate precise grasp poses. Our method addresses these issues by directly leveraging vision-language alignment for grasping, eliminating dependence on external object detectors. By integrating cross-attention with grasp queries, our approach accurately localizes targets even in occluded and cluttered environments. As a result, it achieves higher grasp success rates and improved robustness.

4.5. Ablation study

To evaluate the contribution of each network component, we conduct ablation experiments on four core modules: the CLIP-Adapter, k-NN transformer, multi-scale point cloud fusion, and LGFA module. Moreover, we perform ablation study of the residual ratio α. The results of the module ablations are reported in Table 1. Removing the LGFA module causes a marked drop in success rates for tasks

involving semantically similar objects, (e.g., distinguishing a "yellow mango" from a "yellow lemon") due to reduced cross-modal matching precision. Eliminating the k-NN module, severely impacts performance in highly cluttered scenes where the target is partially or fully occluded, as spatial relationship reasoning becomes unreliable. The absence of the multi-scale point cloud fusion module leads to substantial degradation in scenarios with small or irregularly shaped objects, such as a "green tennis ball" in dense clutter where precise grasp pose estimation is essential for success.

The ablation results for the learnable parameter α in the CLIP-Adapter are summarized in Table 6. From Table 6, we can see that the best residual ratio of our dataset is 0.6. Note that when α equals to 0, it is equivalent to CLIP since no new knowledge is learned. When α is set to 1.0, the classification is fully relies on the adapted feature. However, this is not optimal because CLIP-Adapter tends to over-fit in such condition. The residual connections in CLIP-Adapter offer key advantages for our open-vocabulary grasping task: They prevent over-fitting to limited training examples by retaining pretrained knowledge, critical for recognizing unseen objects in cluttered scenes.

5. Discussion

Strengths Firstly, our method does not rely on explicit visual grounding to identify target objects, thereby avoiding failures caused by incorrect grounding results. By jointly modeling vision, language, and action in an end-to-end manner, the system directly generates grasp poses guided by language, which significantly enhances robustness and reduces error accumulation across modules. Secondly, the integration of multi-scale voxel point cloud fusion improves spatial representation, enabling accurate and efficient grasp planning in cluttered and unstructured environments. Furthermore, our approach preserves the accuracy advantages of two-stage methods while achieving high efficiency and real-time performance through modular design, directly aligning with the practical demands of robotic applications. We regard end-to-end models as the future direction for robotic grasping, and our method not only advances performance in this domain but also provides a forward-looking framework for future developments.

Limitations Firstly, the current system adopts a two-finger rigid gripper, which is inherently limited in handling delicate or deformable objects such as strawberries or tomatoes. While grasping force can be reduced to avoid damage, rigid contact still poses a significant risk of deforming or crushing such items, making it unsuitable for applications involving fragile materials. A potential solution would be to integrate a soft or multi-fingered gripper, which can adapt to object shape and distribute force more evenly. Secondly, our method is less effective when dealing with objects that have highly irregular or complex geometric structures. This is mainly because we currently select grasp poses based on the Euclidean distance between the grasp pose center and the bounding box center, which has limitations. For such objects, the centroid often deviates significantly from the bounding box center. To address these issues, we plan to introduce geometric constraints in future work. We will incorporate surface normals and curvature to prioritize stable regions. We also aim to integrate collision detection modules to filter out poses that are prone to collisions. We believe these steps will enhance robustness in complex scenes. Thirdly, the system remains vulnerable to variations in lighting conditions and the presence of transparent objects. These factors reduce the contrast between the target and its background, often leading to unreliable or failed object detection. As accurate perception forms the basis for downstream grasp pose estimation, such degradations significantly compromise overall grasping performance. We will continue to address the above limitations in our future research.

6. Conclusion

In this paper, we present a novel approach to language-guided open-vocabulary grasping in cluttered environments. Specifically, we propose a visual-linguistic perception system to achieve accurate object localization and successful grasping of novel objects by joint visual-language features and spatial features. We evaluate our method through a series of simulation and real-world experiments in our experiments, which indicates that our method can achieve better task success rate with fewer grasps than other methods under open vocabulary. In addition, our method can also be used for unseen language templates. We verify the effectiveness and robustness of Ours in experiments by comparing it with several classical and advanced approaches in both synthetic and real scenes.

CRediT authorship contribution statement

Xiaomei Zhang: Validation, Supervision, Investigation. **Hanyue Ling:** Writing – original draft, Visualization, Methodology, Data curation. **Xiao Huang:** Validation, Supervision, Conceptualization. **Qiwen Jin:** Supervision, Methodology. **Jiwei Hu:** Supervision, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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